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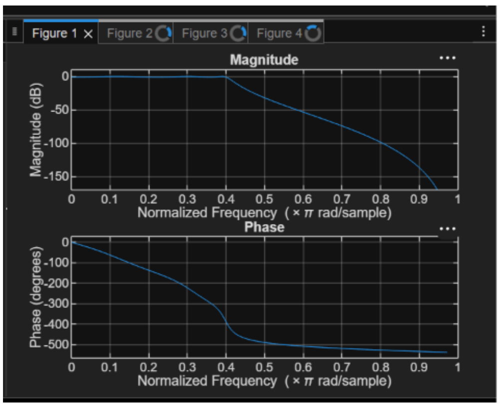
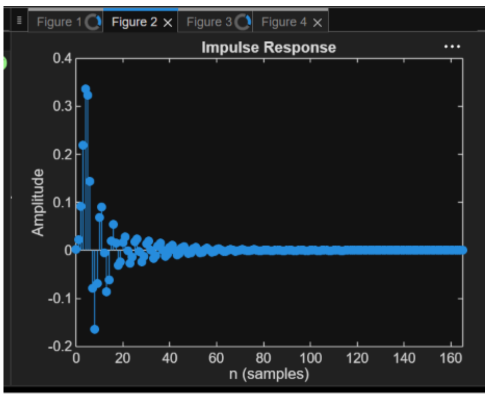
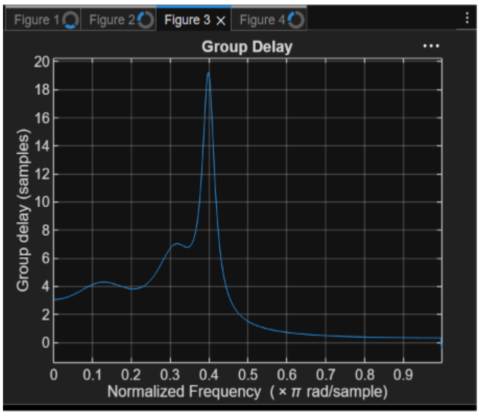
EXP: 2.1:

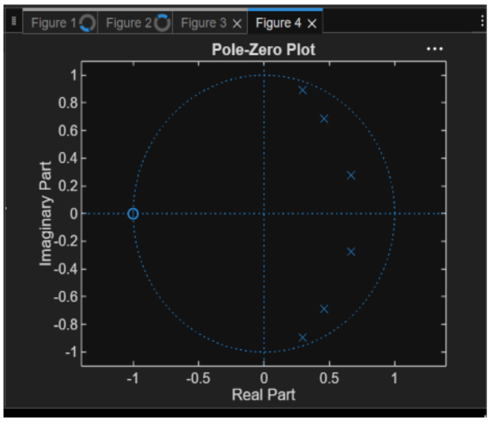
```
clc; clear;

Wp=0.4; Ws=0.55; Rp=1; As=40;

[N,Wn]=cheb1ord(Wp,Ws,Rp,As);

[b,a]=cheby1(N,Rp,Wn);

fprintf('Filter Order = %d\n',N);

fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);

if all(abs(roots(a))<1)

   disp('System is STABLE');

else

   disp('System is UNSTABLE');

end

% --------- GRAPHS ---------

figure; freqz(b,a); % Magnitude + Phase

figure; impz(b,a); % Impulse Response

figure; grpdelay(b,a); % Group Delay

figure; zplane(b,a); % Pole-Zero
```

OUTPUT:









EXP:2.2:

CODE:

```
clc;
clear;
close all;
% Specifications
Rp = 1; % Passband Ripple (dB)
Rs = 40; % Stopband Attenuation (dB)
Wp = 0.5; % Passband Edge
Ws = 0.35; % Stopband Edge
% Find Filter Order
[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);
% Design High-Pass Filter
[b,a] = cheby1(n,Rp,Wn,'high');
% Frequency Response
figure;
freqz(b,a);
title('Frequency Response');
% Impulse Response
figure;
impz(b,a);
title('Impulse Response');
% Group Delay
```

```
figure;
grpdelay( b,a);
title('Group Delay');
% Pole-Zero Plot
figure;
zplane(b,a);
title('Pole-Zero Plot');
fprintf('Filter Order = %d\n',n);
fprintf('Cutoff Frequency = %.4f\n',Wn);
```

OUTPUT: